

Solution to Ramanujan Challenge Problem 2.1: Polynomial Continued Fraction for $6/(3 - \pi)$

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Abstract

We prove that the polynomial continued fraction (PCF) with $a_n = -220n^3 - 484n^2 - 301n - 42$ and $b_n = 4n^2(2n + 1)^2(5n - 4)(5n + 6)$ converges to $6/(3 - \pi)$. The proof reduces to a known continued fraction for π from Cohen's database (arXiv:2409.06086, Entry 1.2.18) via an elementary sign-flip identity on convergents.

1 Statement

Define

$$a_n = -220n^3 - 484n^2 - 301n - 42, \quad (1)$$

$$b_n = 4n^2(2n + 1)^2(5n - 4)(5n + 6). \quad (2)$$

We prove:

$$a_0 + \frac{b_1}{a_1 + \frac{b_2}{a_2 + \frac{b_3}{a_3 + \cdots}}} = \frac{6}{3 - \pi}. \quad (3)$$

2 Cohen's continued fraction for π

Henri Cohen's encyclopedic dictionary of polynomial continued fractions [3] contains Entry 5.3.22 (numbered 1.2.18 in the earlier database [1]):

$$\pi = 3 + \frac{6}{\alpha(1) + \frac{\beta(1)}{\alpha(2) + \frac{\beta(2)}{\alpha(3) + \cdots}}} \quad (4)$$

where, for $n \geq 1$,

$$\alpha(n) = 220n^3 - 176n^2 - 7n + 5, \quad (5)$$

$$\beta(n) = 4n^2(2n + 1)^2(5n - 4)(5n + 6). \quad (6)$$

This continued fraction converges to π at an exponential rate of $\varphi^{-10} \approx 10^{-2.09}$ digits per term, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio.

2.1 Rigorous derivation of (4)

Entry 5.3.22 is not merely a numerical observation: it follows from a rigorous chain of rational identities.

Step 1: Hypergeometric seed (Entry 5.3.11). Cohen's Entry 5.3.11 gives the continued fraction

$$\pi = 3 + \frac{3}{\tilde{\alpha}(1) + \frac{\tilde{\beta}(1)}{\tilde{\alpha}(2) + \dots}}, \quad (7)$$

with $\tilde{\alpha}(n) = 8n^2 + 4n - 1$ (and $\tilde{\alpha}(1) = 11$) and $\tilde{\beta}(n) = -8n(2n+1)^3$. This seed is proved by the elementary identity $\pi = 6 \arcsin(1/2)$: the Euler transformation of $\arcsin(1/2) = \sum_{n=0}^{\infty} \frac{1}{2^{2n+1}(2n+1)} \binom{2n}{n}$ into a continued fraction yields (7) exactly.

Step 2: Apéry acceleration (Entry 5.3.16). Cohen [2] applies Bauer–Muir transformations with an explicit rational simplifier to the seed (7). The resulting bivariate continuant arrays satisfy exact rational identities, and the staircase path through them produces the period-two continued fraction Entry 5.3.16, whose convergents are Apéry-accelerated convergents of (7) converging to π at the golden-ratio rate φ^{-5} per half-step.

Step 3: Even contraction to 5.3.22. The standard even-contraction formula applied to the period-two fraction 5.3.16, followed by a polynomial equivalence transformation with $r_n = \frac{(5n+1)(5n-4)}{n(4n-1)}$, yields (4) with coefficients (5)–(6). This is a finite algebraic certificate independent of any numerical testing.

3 Polynomial identities

Lemma 1 (Index shift). *For all $n \geq 0$,*

$$a_n = -\alpha(n+1) \quad \text{and} \quad b_n = \beta(n).$$

Proof. For the first identity:

$$\begin{aligned} \alpha(n+1) &= 220(n+1)^3 - 176(n+1)^2 - 7(n+1) + 5 \\ &= 220n^3 + 660n^2 + 660n + 220 - 176n^2 - 352n - 176 - 7n - 7 + 5 \\ &= 220n^3 + 484n^2 + 301n + 42. \end{aligned}$$

Hence $-\alpha(n+1) = -220n^3 - 484n^2 - 301n - 42 = a_n$.

The second identity is immediate: both b_n and $\beta(n)$ are $4n^2(2n+1)^2(5n-4)(5n+6)$. □

4 Sign-flip lemma

Lemma 2 (Sign-flip identity for convergents). *Let $\{c_k\}_{k \geq 0}$ and $\{d_k\}_{k \geq 1}$ be sequences with $d_k \neq 0$, and define the convergents of the continued fraction $S = c_0 + d_1/(c_1 + d_2/(c_2 + \dots))$ by the recurrence*

$$X_n = c_n X_{n-1} + d_n X_{n-2}, \quad n \geq 1, \quad (8)$$

with $P_{-1} = 1$, $P_0 = c_0$, $Q_{-1} = 0$, $Q_0 = 1$.

Define the “sign-flipped” continued fraction $\tilde{S} = -c_0 + d_1/(-c_1 + d_2/(-c_2 + \dots))$, whose convergents satisfy

$$\tilde{X}_n = -c_n \tilde{X}_{n-1} + d_n \tilde{X}_{n-2}, \quad n \geq 1, \quad (9)$$

with $\tilde{P}_{-1} = 1$, $\tilde{P}_0 = -c_0$, $\tilde{Q}_{-1} = 0$, $\tilde{Q}_0 = 1$.

Then for all $n \geq -1$:

$$\tilde{P}_n = (-1)^{n+1} P_n \quad \text{and} \quad \tilde{Q}_n = (-1)^n Q_n. \quad (10)$$

In particular, $\tilde{P}_n/\tilde{Q}_n = -P_n/Q_n$ for all $n \geq 0$, so

$$\tilde{S} = -S \quad (11)$$

whenever S converges.

Proof. We prove (10) by induction on n .

Base cases. $\tilde{P}_{-1} = 1 = (-1)^0 \cdot 1 = (-1)^0 P_{-1}$. ✓

$\tilde{P}_0 = -c_0 = (-1)^1 c_0 = (-1)^1 P_0$. ✓

$\tilde{Q}_{-1} = 0 = (-1)^{-1} \cdot 0 = (-1)^{-1} Q_{-1}$. ✓

$\tilde{Q}_0 = 1 = (-1)^0 \cdot 1 = (-1)^0 Q_0$. ✓

Inductive step. Suppose $\tilde{P}_{n-1} = (-1)^n P_{n-1}$ and $\tilde{P}_{n-2} = (-1)^{n-1} P_{n-2}$. Then:

$$\begin{aligned} \tilde{P}_n &= (-c_n) \tilde{P}_{n-1} + d_n \tilde{P}_{n-2} \\ &= (-c_n)(-1)^n P_{n-1} + d_n (-1)^{n-1} P_{n-2} \\ &= (-1)^{n+1} c_n P_{n-1} + (-1)^{n+1} (-1)^{-2} d_n P_{n-2} \\ &= (-1)^{n+1} (c_n P_{n-1} + d_n P_{n-2}) \\ &= (-1)^{n+1} P_n. \end{aligned}$$

In the third line we used $(-1)^{n-1} = (-1)^{n+1}$ since $(-1)^{-2} = 1$. The induction for $\tilde{Q}_n$ is identical (with $\tilde{Q}_{-1} = Q_{-1} = 0$). $\square$

5 Proof of the main identity

Theorem 3. *The continued fraction (3) converges to $6/(3 - \pi)$.*

Proof. Let T denote the first tail of Cohen’s continued fraction (4):

$$T = \alpha(1) + \frac{\beta(1)}{\alpha(2) + \frac{\beta(2)}{\alpha(3) + \dots}}.$$

From (4), $\pi = 3 + 6/T$, hence

$$T = \frac{6}{\pi - 3}. \quad (12)$$

By Lemma 1, our challenge PCF has partial quotients $a_k = -\alpha(k+1)$ and partial numerators $b_k = \beta(k)$ for $k \geq 1$, so it equals

$$-\alpha(1) + \frac{\beta(1)}{-\alpha(2) + \frac{\beta(2)}{-\alpha(3) + \dots}}.$$

This is exactly the sign-flipped version of T (Definition of Lemma 2, with $c_k = \alpha(k+1)$ and $d_k = \beta(k)$). By (11),

$$a_0 + \frac{b_1}{a_1 + \frac{b_2}{a_2 + \cdots}} = -T = -\frac{6}{\pi - 3} = \frac{6}{3 - \pi}. \quad \square$$

6 Numerical verification

We have independently verified the identity (3) and Cohen’s base identity (4) to over 1000 decimal digits using arbitrary-precision arithmetic (Python `mpmath`). With N terms computed:

- Cohen’s CF (4) matches π to $\lfloor 2.09N \rfloor$ digits, consistent with Poincaré roots φ^5 and $-\varphi^{-5}$ (convergence base φ^{-10}).
- Our PCF (3) converges at the same rate, as the sign-flip lemma preserves convergent magnitude.

7 Remarks

1. The Poincaré polynomial of the three-term recurrence (8) is $c^2 - 11c - 1 = 0$, with roots $\varphi^5 = (11 + 5\sqrt{5})/2$ and $-\varphi^{-5} = (-11 + 5\sqrt{5})/2$.
2. Cohen’s CF is listed as Entry 1.2.18 in the 2024 database [1] and as Entry 5.3.22 in the 2026 dictionary [3]. The 2026 dictionary identifies it as the even contraction of the period-two fraction 5.3.16.
3. The partial numerator polynomial $\beta(n) = 4n^2(2n+1)^2(5n-4)(5n+6)$ factors into $\binom{2n}{n}$ -related Pochhammer products. The cubic Pochhammer gauge $g_n = \frac{(6/5)_n(4/5)_n(1)_n}{(n!)^2}$ transforms the continuant recurrence into a degree-3 Apéry-type recurrence (cf. [3], Section 5.3).

References

- [1] H. Cohen, *A Database of Continued Fractions of Polynomial Type*, arXiv:2409.06086, 2024.
- [2] H. Cohen, *Apéry Acceleration of Continued Fractions*, 2024. The Bauer–Muir formulas, simplifier, and staircase construction producing period-two π -fractions from the arcsine seed.
- [3] H. Cohen, *Continued Fractions of Polynomial Type: Theory and Encyclopedic Dictionary*, arXiv:2607.06581, 2026.
- [4] Ramanujan Machine Group, *The Ramanujan Challenge for AI*, 2026.